

ID: 24241132

Packet Size: 2424

```
zorox@zorox-VirtualBox:~/ns-allinone-3.43/ns-3.43$ ./ns3 run scratch/fir
st.cc
[0/2] Re-checking globbed directories...
[2/2] Linking CXX executable /home...build/scratch/ns3.43-first-default
At time +2s client sent 2424 bytes to 10.1.1.2 port 9
At time +2.00596s server received 2424 bytes from 10.1.1.1 port 49153
At time +2.00596s server sent 2424 bytes to 10.1.1.1 port 49153
At time +2.01192s client received 2424 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
  Tx Bytes: 2452
  Rx Bytes: 2452
  Mean Delay: 0.0059616 s
  Throughput: 1089.78 bps
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
  Tx Bytes: 2452
  Rx Bytes: 2452
  Mean Delay: 0.0059616 s
  Throughput: 1089.78 bps
```

Packet Size: 1132

```
zorox@zorox-VirtualBox:~/ns-allinone-3.43/ns-3.43$ ./ns3 run scratch/fir
st.cc
[0/2] Re-checking globbed directories...
[2/2] Linking CXX executable /home...build/scratch/ns3.43-first-default
At time +2s client sent 1132 bytes to 10.1.1.2 port 9
At time +2.00386s server received 1132 bytes from 10.1.1.1 port 49153
At time +2.00386s server sent 1132 bytes to 10.1.1.1 port 49153
At time +2.00772s client received 1132 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
  Tx Bytes: 1160
  Rx Bytes: 1160
  Mean Delay: 0.0038592 s
  Throughput: 515.556 bps
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
  Tx Bytes: 1160
  Rx Bytes: 1160
  Mean Delay: 0.0038592 s
  Throughput: 515.556 bps
```

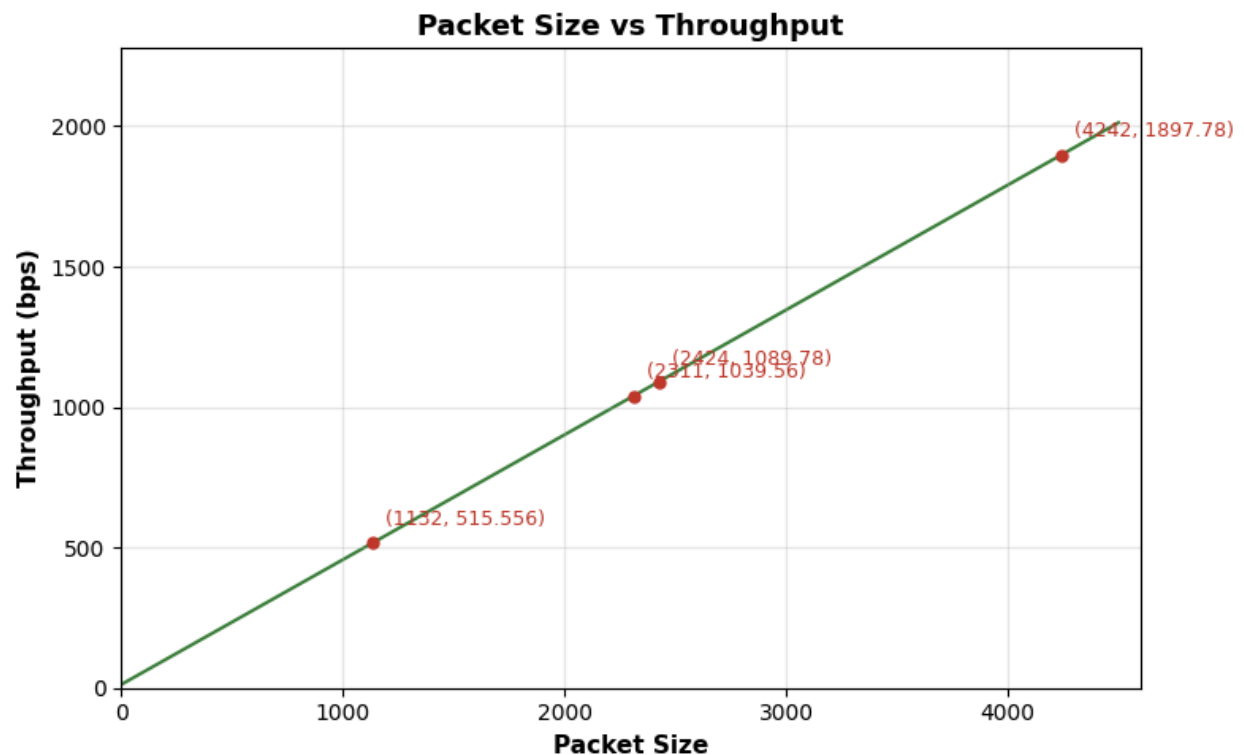
Packet Size: 4242

```
zorox@zorox-VirtualBox:~/ns-allinone-3.43/ns-3.43$ ./ns3 run scratch/fir
st.cc
[0/2] Re-checking globbed directories...
[2/2] Linking CXX executable /home...build/scratch/ns3.43-first-default
At time +2s client sent 4242 bytes to 10.1.1.2 port 9
At time +2.00891s server received 4242 bytes from 10.1.1.1 port 49153
At time +2.00891s server sent 4242 bytes to 10.1.1.1 port 49153
At time +2.01781s client received 4242 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
  Tx Bytes: 4270
  Rx Bytes: 4270
  Mean Delay: 0.0089056 s
  Throughput: 1897.78 bps
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
  Tx Bytes: 4270
  Rx Bytes: 4270
  Mean Delay: 0.0089056 s
  Throughput: 1897.78 bps
```

Packet Size: 2311

```
zorox@zorox-VirtualBox:~/ns-allinone-3.43/ns-3.43$ ./ns3 run scratch/fir
st.cc
[0/2] Re-checking globbed directories...
[2/2] Linking CXX executable /home...build/scratch/ns3.43-first-default
At time +2s client sent 2311 bytes to 10.1.1.2 port 9
At time +2.00578s server received 2311 bytes from 10.1.1.1 port 49153
At time +2.00578s server sent 2311 bytes to 10.1.1.1 port 49153
At time +2.01156s client received 2311 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
  Tx Bytes: 2339
  Rx Bytes: 2339
  Mean Delay: 0.0057808 s
  Throughput: 1039.56 bps
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
  Tx Bytes: 2339
  Rx Bytes: 2339
  Mean Delay: 0.0057808 s
  Throughput: 1039.56 bps
```

Throughput VS Packet Size Graph



Explanation

The graph shows that throughput increases as the packet size increases, and the relationship is approximately linear across the tested range (1132 to 4242 bytes). Throughput increases roughly linearly with packet size. FlowMonitor calculates throughput from the total bytes received over a fixed simulation interval, so as packet size grows, more bits are delivered in that same interval, and the measured throughput scales up almost proportionally, since per-packet overhead stays nearly constant.